EXPERIMENT 3

Student Name: Aryan Yadav **UID: 24BCF10079**
Branch: AIT-CSE (FSD) **Section/Group: 24AIT_KRG2**
Semester: 4
Subject Name: Database Management System **Subject Code: 24CSH-298**

Experiment 3 – PL/SQL Salary and Bonus Calculation

Experiment

Experiment 3: Calculating employee salary and applying a bonus using PL/SQL. This experiment demonstrates variable declaration, arithmetic operations, and displaying output using DBMS_OUTPUT.PUT_LINE.

Aim

The aim of this experiment is to practice writing PL/SQL blocks that perform calculations on employee data, calculate bonuses, and display results in a structured format.

Objective

- To declare variables in PL/SQL.
- To perform arithmetic calculations on employee salary.
- To calculate a bonus amount based on a percentage of salary.
- To display employee details and salary information before and after applying the bonus.

Software Requirements

- Database: Oracle XE or Oracle Live SQL

Practical / Experiment Steps

1. Declare variables for employee ID, name, and salary.
2. Calculate a 10% bonus on the employee salary.
3. Calculate the total salary after adding the bonus.
4. Display employee details and salary information before and after bonus.
5. Take screenshots of outputs for documentation.

EXPERIMENT 3

Procedure of the Experiment

1. Start the system and log in.
2. Open Oracle XE or Live SQL.
3. Connect to the database.
4. Declare variables for employee details and bonus calculation.
5. Write the PL/SQL block to calculate bonus and total salary.
6. Execute the PL/SQL block and verify outputs in the console.
7. Take screenshots of the outputs (s1 and s2).

Input / Output Details

Input

- Employee details:
  - emp_id NUMBER
  - emp_name VARCHAR2(50)
  - emp_salary NUMBER
- Bonus calculation: 10% of employee salary

Output

- Step 1: Display employee details and original salary.
- Step 2: Display bonus amount and total salary after applying 10% bonus.
- Screenshots of outputs (s1 and s2) are attached.

Step 1 Output

```
Employee Details
----------------
Employee ID   : 121
Employee Name : Aryan

Salary Details
----------------
Salary Before Bonus : 50000
10% Bonus Amount    : 5000
Salary After Bonus  : 55000
```

EXPERIMENT 3

Employee Details

Employee ID : 121
Employee Name : Aryan
Salary Before Bonus : 50000

Step 2 Output
Screenshot: Step 2 – Salary After Bonus

Salary Details

Salary Before Bonus : 50000
10% Bonus Amount : 5000
Salary After Bonus : 55000

Learning Outcome
After completing this experiment, the student will be able to:

- Declare and use variables in PL/SQL.
- Perform arithmetic operations for salary calculations.
- Calculate bonuses based on a percentage of salary.
- Display structured outputs using DBMS_OUTPUT.PUT_LINE.
- Understand the workflow of PL/SQL blocks for practical data operations.